

DATA PRIVACY

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Lab 1: Machine Learning Security and Privacy Attacks

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1 Motivation

This lab aims to provide hands-on experience with security and privacy attacks against machine learning systems. The lab contains two membership inference attack scenarios and one data-poisoning backdoor attack scenario.

The goal is to understand the attack process, the assumptions made about the attacker, and how the success of each attack can be evaluated in practice.

2 Lab Instructions

Using the relevant links, you can access the Google Collab Jupiter notebook compiler for each attack. Please read the instructions and execute the program cells one by one.

2.1 Backdoors

In this exercise, you will study a data-poisoning backdoor attack on a text classification system.

You will first train and evaluate a clean spam classifier. Next, part of the training data will be poisoned by introducing a trigger, and the compromised classifier will be evaluated on both clean and triggered inputs. Finally, you will experiment with different triggers and poisoning settings to observe their effect on the Attack Success Rate (ASR).

- [Backdoor Attack](#)
- [Train Data](#)
- [Test Data](#)

2.2 MIA

In the second part of the lab, you will study membership inference attacks (MIAs) against machine learning classifiers. You will investigate two scenarios:

- Simple black-box membership inference, using model correctness, confidence, and loss as membership signals.
- Shadow model membership inference, where an attacker uses auxiliary data and a shadow model to train a separate membership attack classifier.

You will compare the behavior of the target model on members and non-members and evaluate privacy leakage using metrics such as ROC AUC.

- [MI Attacks](#)